Equilibrium Tape Synth (ETS)

A fixed free-energy model for controllable, provenance-preserving corpus-based synthesis

ETS Project
(build documentation; author line placeholder)

Post-v1 documentation

Abstract

The Equilibrium Tape Synth (ETS) is a musical instrument that a performer plays live. Instead of generating audio from scratch, it continuously rearranges small fragments of real recordings into an output "tape," steered in real time by a handful of controls, and every sound it produces can be traced back to the exact recording it came from. Under the hood, ETS is built around a single learned energy function whose lowest-energy arrangements are real pieces of music; playing the instrument amounts to letting each new bar settle toward one of those low-energy arrangements while the controls gently lean the result one way or another. A second theme of the paper is how the system was built: a discipline for keeping the software honest to its written specification, using automated checks that must themselves be shown to catch deliberate mistakes, an independent adversarial reviewer, and an append-only log of every design decision and experiment. We report what works and what does not — including one central test, of whether the learned model has the geometric structure we had conjectured, that did not pass, and which we report as-is rather than tuning it until it did.

Contributions.

(i) a single free-energy functional F over *unit-resolved* couplings whose low-energy configurations are real tracks, fit by convex logistic NCE against a fixed, pre-registered family of self-generated negatives (no external data);

(ii) an exact, learned-content-free control law — a Gibbs tilt whose per-lane scales are *derived* from equilibrium fluctuations (a fluctuation–dissipation identity), so the six-lane panel adds no free constant;

(iii) a *faithfulness discipline* that is the real deliverable: one authority (F), a two-state (enforced/pending) invariant manifest with checks proven to bite, an adversarial auditor, and an append-only registry that records a caught fidelity breach and a failed headline gate as first-class results.

Authorities. `ets-spec-v0.md` (functional §5, invariants §14, panel §8, render §11); `ets-connector-v0.md` (Layer-0 tilt map); `REGISTRY.jsonl` (build history, honest numbers). Where those and this paper disagree, they win. Every figure below is traced to a committed source or results artifact.

1 Architecture overview

ETS is a pipeline from real audio to a committed output tape. Each stage has a concrete job and a formal object, and the pipeline is organized so that exactly one stage is a decision — the minimization of F — while every other stage is deterministic mechanism, a read-only instrument, or a pure application of an already-decided result. Figure 1 shows the runtime as a single data-flow diagram; the stages, in order, are: ingestion (beat clock, filterbank, unitization) → the fixed model

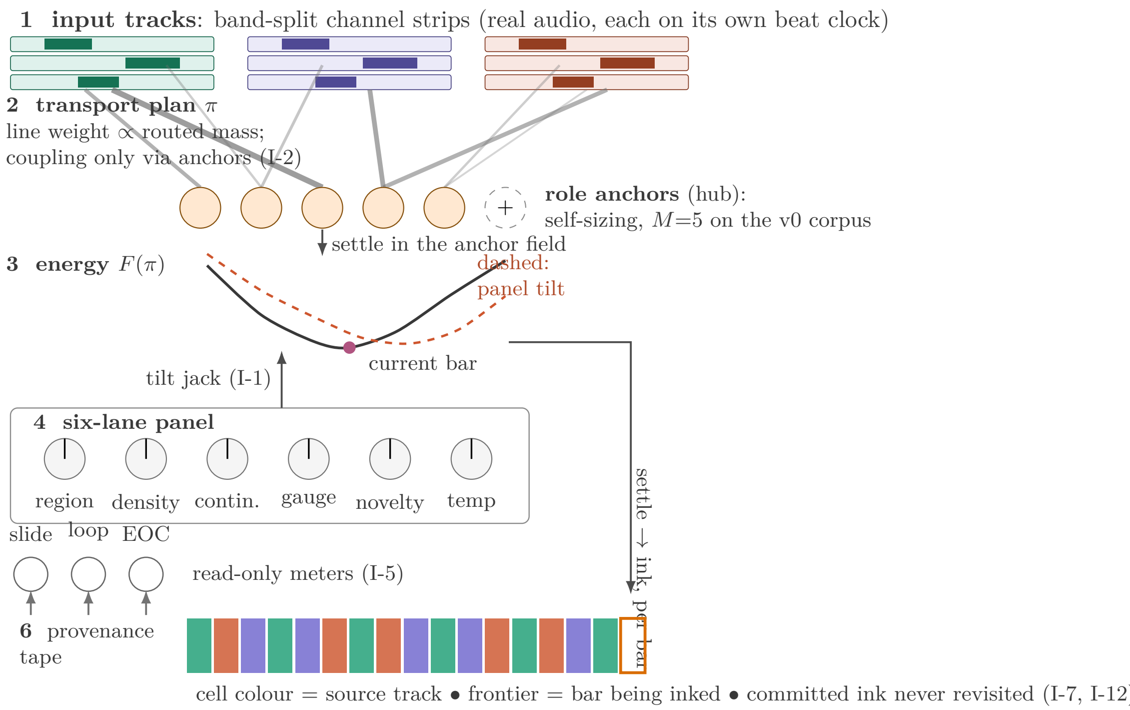

Figure 1: The ETS runtime as the machine's own data flow (a redraw of the project's schematic with its informal labels made precise). **(1)** N input tracks, each a stack of band-split channel strips of real audio on its own beat clock. **(2)** The transport plan π (line weight proportional to routed coupling mass) connects units to a small set of self-sizing role anchors; no track couples to another except through the anchors (I-2), and the anchor count is derived, not set (M=5 on the v0 corpus; the dashed slot marks the self-sizing law). **(3)** The free energy $F(\pi)$; the marked point is the coupling for the current bar settling to its minimum, and the dashed curve is F after the panel's tilt. **(4)** The six control lanes enter the settlement through a single tilt jack (I-1): a knob reshapes F, it does not override the settled result. Each bar is settled and then inked onto the **(6)** provenance tape, whose cell colour records the source track (I-12) and whose committed cells are never revisited (I-7). The read-only meters *slide*, *loop*, and *EOC* report upward from the tape (I-5). Term map: bowl/landscape = $F(\pi)$; marked point = the current-bar coupling; fan = the transport plan π; hub = the role anchors. At v1, *slide* and *loop* are the split two-jack pair; the single conflated drift jack of the original schematic was deleted in the Stage-0/1 meter split.

(self-sizing anchors) → F and its settlement → the writer/tape → rendering, with an optional external mastering stage downstream.

There are two orthogonal control paths, and only two: (i) the six **CV lanes** enter the writer through a *single jack*, the h-transform tilt (I-1); (ii) **clamps** — committed tape, "this unit at bar 33", and knob leans as clamped role-space mass — are all the same boundary-condition species (I-7, generalized by the connector). No third intervention mechanism exists. The desktop runtime is two processes (§4): an ENGINE (Python core, streaming writer, audio) and a PANEL (PySide6/Qt) over OSC on localhost; killing the panel does not affect the engine, and any OSC/MIDI controller can replace the panel.

2 Ingestion, gauge, and the fixed model

Ingestion. Audio is resampled to 44.1 kHz, beat-tracked (`beat_this==1.1.0`) to a tatum/beat/bar grid, and decomposed by a fixed 8-band log-spaced filterbank — deliberately *not* a separator, because "a separator is a second authority pre-deciding roles before F does" (spec §2.3). The raised-cosine masks form a *partition of unity* (`ingestion/filterbank.py`: `N_FFT=2048`, `HOP=512`, `FMIN=40 Hz`); the per-band ISTFTs sum back to the input to numerical precision, which makes the G0 reconstruction identity a coverage/scheduling test rather than a lossy-model test.

Gauge (spec §3). Per track the gauge group is transposition × beat-phase × loudness × timbre-basis. All cross-track communication is invariant under independent per-track gauge action; *no coordinate crosses a track boundary* (I-2: descriptor arrays are private, the sole combiner refuses a cross-track pair, pitch-class cost is transposition-quotiented, metrical cost is circular). Microtiming deviation is *not* gauge — it is intrinsic groove content that must survive, and it is exactly what T_1's phase charge later prices.

The fixed model (self-sizing anchors, spec §4). We call the inference-time model *fixed* in the precise sense that, after training, the role anchors and their pairwise cost matrix D, their masses a, the channel gains B, the metrical-slot targets θ, and the calibrated weights `LAMBDA` are all held constant; nothing in the runtime updates them (the spec calls this the world-freeze). All cross-track traffic factors through free-support barycenter *anchors* in role space; there is no direct pairwise track coupling. Roles are compared across tracks by a Gromov–Wasserstein distance [6, 5], which is defined between metric-measure spaces and so needs no shared coordinate system — exactly the property I-2 requires. The anchor count is not a hyperparameter — it is *derived* as the balanced-truncation effective rank (participation ratio $(\sum w)^2/\sum w^2$) of the cross-track GW role-affinity operator $e^{-D_{\mathrm{role}}/\sigma}$ (`anchors.py:effective_rank`). On the v0 world this is $M = 5$ (eff_rank ≈ 5.354, $\sigma \approx 0.553$). The supports (D, a, B, θ) are then settled as the GW barycenter at that count. Anchors hold *only* support and mass — no accumulator, EMA, or momentum (I-3, structurally scanned).

Note. The effective-rank rule is a *re-derivation after a wall*: the first sizing rule (a GW transport-residual growth criterion) provably failed to dissociate role-diverse from same-role corpora and grew with N, and was rejected; the current rule reads the spec's own "McMillan degree of *traffic* / Hankel mass / balanced truncation" language as a spectral rank (`RETRO_AUDIT.md`). The wall was reported and F was re-posed, not patched.

3 The energy-based model

The central object is a single free-energy functional F; the system has no training loss distinct from it (invariant I-4). ETS is therefore an energy-based model [1, 2] in the standard sense: F is the energy, a scalar temperature T_s sets the sampling width, and control is an exponential tilt in a set of sufficient statistics φ. Real tracks are the low-energy configurations of F; their rearrangements are not.

3.1 F over the unit-resolved coupling

F ranges over couplings π (unit → role → metrical slot), channel gains B, anchor supports/masses (D, a), and gauge sections g. Crucially, F *is posed on the full unit-resolved* π, not on its role-occupancy marginal $O[k, s] = \sum_{\mathrm{units}} \pi$. A term may use the marginal *only where it provably factors through it*; a term that needs the fiber (which real unit sits where) reads π directly (Table 1).

Table 1: The five terms of F (`functional/f.py`, rev-r1).

Term	Meaning	Posed on	Reads
T_1	GW transport *plus* a circular metrical phase-displacement charge	full-π	GW via prototypes; charge via the unit fiber
T_2	mass conservation (unbalanced-OT gen.-KL of O vs $a\theta$)	marginal O	written factorization proof
T_3	spectral masking (collision $\sum E^2 - \text{self}$, $E = O^\top B$)	marginal O	written factorization proof
T_4	unit-successor run-continuation (Doob h-transform)	full-π	π's fiber; not expressible over O
T_5	per-*section* gauge-fixing (global transpose/phase), never per-unit	gauge	neither π nor O

The **phase-displacement charge** assigns a cost to metrical displacement (groove). For each placed unit it charges the circular distance between the unit's *intrinsic* metrical coordinate x_u and the phase of the slot it occupies, quotiented by a single per-section global phase shift δ (a global beat-phase shift is free gauge; a per-unit displacement is charged). The δ-quotient is solved in closed form,

$$\min_{\delta} \sum_u m_u\big(1-\cos 2\pi(x_u-\delta)\big) \;=\; 1-\Big|\textstyle\sum_u m_u\, e^{\,i2\pi x_u}\Big| \qquad \text{(masses normalized)}, \tag{1}$$

which is exactly 0 when every unit sits at its own intrinsic slot up to one global shift (a real track's groove) and strictly positive for incoherent displacement. T_4 rewards output-adjacent slots holding genuine *source* successors: a grid-shuffle re-deals content so almost no adjacency survives; a cross-track graft inserts units that are no track's successor, so the run breaks. Both fiber terms are gauge-invariant by construction (I-2). Holonomy, drift, novelty, and any meter appear *nowhere* in F (I-5, I-14; enforced by an AST scan that bites on a `holonomy_drift` identifier and on any `ets.meters` import in the F path).

3.2 The Gibbs measure and the Layer-0 tilt

The equilibrium measure over an arrangement a, with control applied, is

$$p(a) \;\propto\; \exp\Big(-\tfrac{F(a)}{T_s}+\sum_i \lambda_i\,\varphi_i(a)\Big), \qquad \lambda_i=\frac{u_i}{\sigma_{\varphi_i}} \qquad \text{(connector Layer 0)}. \tag{2}$$

Here T_s is the *temperature* (sampling looseness around the settled optimum; it carries no φ); the φ_i are five gauge-invariant arrangement statistics (region, density, continuity, gauge-move magnitude, novelty) computable from the candidate arrangement alone (`connector/phi.py`). The tilt is not a bolt-on: it is the *Doob conditioning* the settlement inherits when the panel's boundary measure is clamped. Two of the five φ (region, density) are linear in O and enter the O-block; continuity and novelty live on the fiber block; gauge on the (v0-frozen) gauge block. Run-time knobs move λ and T_s, never F's term weights (I-9).

3.3 Contrastive (NCE) training and the KILL condition

The weights `LAMBDA` are fit *once*, at corpus time, by a convex logistic NCE [3] (spec §6, `training/nce.py`). The condition is: *each real track is an equilibrium of F; its rearrangements* *are not*. There is *no external "bad music" data anywhere* (I-6): the comparison class is generated internally by disarranging each real track's own units, drawn only from a fixed, pre-registered scramble family (Table 2). Because F is linear in the weights and φ is computed at the frozen `LAMBDA`-free world, the NCE objective is *convex* with an exact gradient — no circularity. The fit uses seeds $\{1, 2, 3\}$; held-out validity is measured on *disjoint* seeds $\{4, 5\}$, so no fit metric is a gate metric (I-5).

Table 2: The fixed pre-registered scramble family (the only negatives).

Scramble	What it breaks	Arity
grid-shuffle	metrical placement (re-deal real units within a band)	Track→Track
phase-rotate	the single gauge-phase frame (incoherent *per-band* rotation)	Track→Track
role-permute	role assignment (derange anchor columns of the pure-GW coupling)	(Track, world)→Arr.
cross-track-swap	anchor-mediated cross-track coherence (swap anchor rows)	(Track^2, world)→Arr.

The KILL, then rev-r1 (the decisive negative result). The *first* F was posed on the occupancy *marginal* O. It **failed**: on the marginal, F could not separate real from grid-shuffle — held-out separation ≈ 0.35, a *negative margin* — because the marginal is blind to *which* unit sits at *which* slot (registry `train-nce-2026-07-13` KILL). This was ruled a *fidelity breach*: a silent aggregate projection of π, an unfaithful implementation *plus* a missing invariant (registry `fidelity-breach-2026-07-13`). Per the wall protocol it was reported, not patched: no `LAMBDA` was emitted, the referee was left incorruptible, and F was re-posed on the full unit-resolved fiber (rev-r1). A new invariant, **I-15** (no premature aggregation / structure-deleting projection), was added to forbid the class. The re-posed F (fork C: richer fiber + gauge-aligned groove target) **passed** (Table 3; `training_results.json`, registry `train-nce-revr1-2026-07-13`).

Table 3: Held-out per-member separation, rev-r1 fork C (disjoint seeds $\{4, 5\}$).

Family member	held-out separation	median margin
grid-shuffle	**1.00**	18.79
role-permute	**0.95**	1.07
phase-rotate	**1.00**	6.02
cross-track-swap	**0.975**	7.33
overall min	**0.95** ≥ 0.90 (pre-registered)	—

(logistic loss 0.0908, grad-norm 0.0024, 240 pairs.) The emitted weights, now *frozen* and read live by the writer (I-9), are

$$\texttt{LAMBDA} = \{\, T_2 : 4.9923,\ T_3 : 0.8023,\ T_4 : 10.4577,\ T_{1p} : 8.7581,\ T_5 : 0.1 \,\},$$

with T_1's GW transport as the reference scale (weight 1). $T_5 = 0.1$ is a run-time gauge baseline, *not* corpus-identifiable (a global section-gauge move is orthogonal to every rearrangement, so it carries no contrastive signal). An independent audit re-ran the fit and reproduced `LAMBDA` bit-exactly to 16 digits.

3.4 Settlement as certified I-projections

Minimization is block-coordinate I-projections (`functional/solver.py`): Sinkhorn [4] on π-blocks, exponentiated-gradient (mirror descent) on B-blocks, unbalanced multiplicative updates on masses, a closed-form Gromov–Wasserstein barycenter [7] on D, exact minimization on the gauge block. The certificate is a *Lyapunov F-descent guard*: every block step is accepted *only* if F does not increase, so the F-trajectory is monotone non-increasing and the run stops at $|\Delta F| <$ tol. The accept/reject decision reads F and nothing else — no second objective, no meter (I-4/I-14, verified by parsing `batch_solve` and confirming the guard compares $F_{\text{cand}}/F_{\text{cur}}$). In streaming mode the batch Lyapunov certificate becomes a per-step frontier F-descent plus a *bounded-state* certificate: state growth on stationary input is a broken instrument — halt and report (I-8).

4 The control surface

Six CV lanes, exhaustive (spec §8); a seventh requires a spec revision (Table 4).

Table 4: The six lanes, their φ statistics (`connector/phi.py`), and v0 status.

#	Lane	φ statistic	v0 status
1	REGION TILT (XY / channel strips)	φ_{region}: mass by anchor role	armed (σ per anchor)
2	DENSITY	φ_{density}: bar scheduled mass	**disarmed** (σ=0 at u=0)
3	CONTINUITY ↔ RECOMB.	φ_{cont}: source-successor count	armed ($\sigma \approx 0.301$)
4	GAUGE STIFFNESS	φ_{gauge}: frame-move magnitude	**disarmed** (σ=0, frozen frame)
5	NOVELTY PRESSURE	φ_{novelty}: $1/\Delta$ recency reuse	armed ($\sigma \approx 0.330$)
6	TEMPERATURE	*(no φ)*: settlement sharpness	armed

The knob-scaling law is derived, not hand-set. Each lane's scale is $\lambda_i = u_i/\sigma_{\varphi_i}$, where σ_{φ_i} is the *equilibrium fluctuation* of φ_i under the *untilted* writer, measured by a registered calibration pass at world-freeze (`calibration/sigma_phi.json`). The derivation is fluctuation–dissipation: with $p_u \approx p_0\, e^{\lambda\varphi}$ one has $d\langle\varphi\rangle/d\lambda|_0 = \mathrm{Var}_0(\varphi)$, so $\lambda = u/\sigma$ gives $d\langle\varphi\rangle/du|_0 = \sigma$ — *one knob unit = one equilibrium-σ lean.* σ is the unique normalizer with this property and makes the tilt invariant to any rescaling of φ. No hand-set λ scale exists in the runtime. The calibration ensemble was 7495 bars (the corpus bar count — the evidence scale, not a hand constant), $s_{\text{phase}} = 8$.

Two lanes are disarmed at v0. The untilted writer settles by a deterministic MAP descent on a bar-periodic anchor field, so φ_{density} (an O-marginal) is *pinned* constant ($\sigma = 0$ exactly), and the single identity-gauge section makes $\varphi_{\text{gauge}} \equiv 0$. Rather than invent a floor, the loader marks these lanes `identifiable=false`: u still transmits, but no tilt is applied and the engine surfaces the state (`WorldNotCalibrated` on a lean into an uncalibrated world). This is a theorem about degenerate exponential families reported as a wall, not an error-hiding branch.

Two processes over OSC (spec §12). The ENGINE (`ets/engine`) owns the fixed model, the streaming writer, and audio (`sounddevice`, lazy, headless-graceful). The PANEL (`ets/panel`, PySide6/Qt) is a pure control surface: six lanes, XY vector pad, meter jacks as LEDs, patch-cable feedback metaphor, MIDI CC learn. IPC is OSC over localhost (`python-osc`); no web technology anywhere in the runtime (I-13, enforced by an import scan that even distinguishes the runtime's own `ets.panel` from the forbidden HoloViz `panel`). Control latency is a *declared* L bars measured by buffer math on the host (desktop: $L = 2$ bars, T_{prod} mean 0.568 s $< T_{\text{bar}}$ 1.486 s; `latency_desktop.json`); an underrun inside $L \cdot T_{\text{bar}}$ is a WALL — there is no degraded-quality fallback.

Meter jacks. Read-only, never in any objective/gradient/settlement (I-5). The conflated DRIFT jack was split into two shadow meters — `slide` (per-bar displacement of the settled gauge frame, in F's own quotient) and `loop` (a per-committed-bar transplant of the G2 loop-defect integrator) — then the old conflated jack was *deleted* after a registered one-shot regression showed it is exactly informationless on every producible trace (residual 0.0 at machine precision, registry `conflation-regression-stage1`).

5 The faithfulness discipline

Faithfulness — agreement between the implementation and its specification — is enforced by executable checks and process rules rather than asserted.

Two-state invariant manifest (`tests/invariants/manifest.py`). Every invariant I-1..I-15 carries exactly one of two statuses — ENFORCED (an executable check that raises on violation) or PENDING (the guarded feature does not exist yet). There is *no third state*: PENDING never means "we chose not to check," and no invariant may be absent or vacuously satisfied. When a feature lands, its invariant moves to ENFORCED *in the same change* or the auditor rejects the diff. Current state: **14 of 15 enforced**; only **I-10 (thin planner) is pending**, because the planner is not built. The checks are aggressively non-vacuous — each proves it *bites* against a planted mutant (a choosing render, an accumulator field, an η·KL tether, a web import, an EOC-forked settlement guard). Table 5 lists all fifteen.

Table 5: The 15 invariants and enforcement state (spec §14 + connector I-15).

	Invariant	Enforced
I-1	single tilt jack — no control path but the h-transform tilt	✓
I-2	gauge law — no coordinate crosses a track boundary	✓
I-3	no pressure accumulator / duplicate smoothing	✓
I-4	one F — no training loss distinct from F	✓
I-5	meters never in objective/gradient/settlement	✓
I-6	no external negatives; scramble family fixed in PREREG	✓
I-7	all interventions are clamped cells; no recovery modes	✓
I-8	streaming stability; halt on state growth (stationary input)	✓
I-9	run-time controls are tilt parameters only; F weights frozen	✓
I-10	**planner stateless, external, thin**	**pending**
I-11	rendering applies, never chooses	✓
I-12	provenance — every sample traces to (track, unit, transform)	✓
I-13	no browser/web tech in the runtime	✓
I-14	Hankel/holonomy are instruments; no fork of F's authority	✓
I-15	no premature aggregation / structure-deleting projection	✓

Builder/auditor adversarial pair. Implementation is done by an `ets-builder` agent and must be paired with a read-only `ets-auditor`; no gate runs and no merge happens without an auditor PASS. The auditor reports verdicts and never fixes code.

Prereg-commit-before-run, append-only registry. Every gate's hypothesis, procedure, null, and kill condition is appended to `PREREG.md` and committed *before* the run; `REGISTRY.jsonl` is append-only. Corrections are new entries, never edits.

Operating slogan. *Walls are information, patches are sabotage, silent divergence is the breach class.* A wall (a failed gate, a non-identifiable scale, an unmeetable deadline) is surfaced and owned; the failure mode the discipline exists to catch is a *silent* divergence between what the spec defines and what the code does.

Case study — the fidelity breach (the discipline working)

The most instructive event in the build is a *fidelity breach caught before delivery* (registry `fidelity-breach-2026-07-13`). An early implementation silently collapsed the unit-resolved coupling π to its marginal O — a structure-deleting conversion shim. The spec defines π as unit-resolved (§2/§5); the implementation aggregated without presenting the divergence. It was caught not by luck but by the *pre-registered exam*: the fixed scramble family correctly refused to separate (grid-shuffle held-out ≈ 0.35, negative margin), the KILL condition fired, and — critically — no `LAMBDA` was emitted and the wall was surfaced pre-delivery. Remediation: (1) F re-posed on the full fiber; (2) a new invariant **I-15** with two teeth — a term-input contract in `f.py` behaviorally verified against an O-preserving π rearrangement, and a referee-non-degeneracy check that the scored corpus-time feature *must* consume a unit-resolved fiber. The incident is logged `logged-permanent-no-relitigation`. The episode is evidence that the exam functions as intended: the check designed to catch an unfaithful implementation caught one, and the remediation left a permanent guard (I-15) behind it.

6 Related work

ETS is a *conjunction* of established primitives, and its positioning claim is deliberately narrow: the novelty is the combination and the discipline, not any one mechanism. We state the prior art plainly (mirroring the project's own "occupancy ledger," spec §16).

Corpus-based and concatenative synthesis. Assembling output from real segmented source units, chosen by descriptor proximity, is the concatenative / corpus-based tradition — CataRT and the surrounding survey [9, 10]. ETS shares the "play real grains" commitment but differs in what *chooses*: not nearest-neighbour in a descriptor space but the equilibrium of a single functional, with placement carrying settled mass and full provenance. Mashup systems such as AutoMashUpper [11] pre-decide alignment by harmonic/rhythmic "mashability"; ETS instead routes all cross-track traffic through self-sized anchors and lets F, not a similarity score, set the coupling.

Energy-based models and contrastive estimation. The energy-as-model framing and its training are classical [1, 2]; noise-contrastive estimation [3] is our estimator, specialized so that the negatives are the corpus's *own* disarrangements (no external data, I-6) and the fit is convex in the weights. Unlike a typical deep EBM, F here is a transparent optimal-transport free energy with named terms, and no run-time control edits its weights (I-9).

Optimal transport for audio and music. Sinkhorn-regularized OT [4, 5] and Gromov–Wasserstein matching/averaging [6, 7] are the computational backbone of the world and the settlement. OT has been used directly on audio, e.g. spectral transport for portamento/morphing [8]; ETS uses OT one level up — on beat-clocked *unit clouds* and their role geometry — and turns the transport plan into a schedule of real units rather than an interpolated spectrum.

Neural audio codecs and controllable synthesis. Contemporary systems tokenize audio with learned codecs [13, 14] and generate with language models over those tokens [15], or control synthesis through differentiable DSP [12]. ETS occupies a different point: it has *no decoder and no learned readout* — the tape node's coupling *is* the output and the render only applies it (I-11/I-12) — and its single learned object is the frozen scalar-weight vector `LAMBDA`, everything else being derived mechanism or read-only instrument. The trade is explicit: ETS buys faithfulness and provenance at the cost of the open-ended generative range a codec-plus-LM enjoys.

7 Limitations: what is real versus pending

Table 6 records, for each component, whether it is implemented and verified, a declared approximation, or not yet built, as of v1.

G2 in detail (the dissociation that did not hold). G2 tests whether the fixed model's role geometry carries *holonomy* (loop-defect curvature) above a residual-conditioned null — the dissociation that would support the conjecture that musical structure is real geometric curvature, at a demanding 20× target. It **failed** (`g2_results.json`): real median curvature came in *below* the null ($\kappa_{\text{real}}/\kappa_{\text{null}} \approx 0.45$), dominance over null p95/p99 both false, `signal_present_above_null=false`. Real curvature *did* clear the solver floor ($\kappa_{\text{real}}/\kappa_{\text{floor}} \approx 2.0$), but not the residual-conditioned null. Per the pre-registered kill condition this is reported as-is, with no null-swap and no retune. Two consequences stated plainly: (i) the strong "curvature = structure" claim is *unsupported at v0*; (ii) the instrument survives as a read-only meter (`loop`) regardless. The pairwise-blindness and closed-loop *form* theses (G3–G6) depend on the planner, which is not built (I-10 pending), so they remain untested.

Table 6: Status ledger. Declared approximations and unpassed gates are reported, not rounded up.

Item	Status	Note
Ingestion $\to$ world $\to$ F $\to$ writer $\to$ render	REAL	deterministic pure render (H-8 receipt)
Anchor self-sizing (eff. rank of traffic)	REAL	$M = 5$ on v0 corpus; G1 passed
Contrastive NCE fit of `LAMBDA` (rev-r1)	REAL	min held-out sep $0.95 \geq 0.90$; audit-reproduced bit-exactly
14/15 invariants enforced	REAL	only I-10 pending
Two-process engine+panel over OSC, MIDI CC learn	REAL	headless-graceful; $L = 2$ bars declared
G0 ingestion/reconstruction	**pass**	recon rel-L2 $\approx 1.16\times10^{-8}$
G1 anchor double dissociation	**pass**	diversity gap 2.04; SAME 2.38 $<$ NULL 3.30 $<$ DIVERSE 4.43; flat-in-N
G2 holonomy above calibrated null	**did not pass**	$\kappa_{\text{real}}/\kappa_{\text{null}} \approx 0.45$; reported as-is, no retune
REGION / CONTINUITY / NOVELTY / TEMPERATURE lanes	ARMED	derived σ_φ scales
DENSITY / GAUGE lanes	DISARMED	$\sigma_\varphi = 0$ at u=0; λ undefined, no tilt, surfaced
Greedy fiber realization	DECLARED APPROX	uncertified vs a joint fiber settlement; PASS-legal with declaration
Streaming temperature sampler	DECLARED APPROX	per-slot Laplace 2nd-order + orthant clip ($\sim$ 0.11 std bias, $\sim$ 13% clip at T_s=1)
mass = $\sqrt{e[b]}$ settled-mass render	DECLARED APPROX	exact on schedule; adjacent-band crossover leakage ($\sim$ +0.15–0.20 worst case)
I-10 planner	**pending**	closed-loop form thesis (G3–G6) untested
Mono	DECLARED V0	stereo restoration is real future scope, not a patch
External output master	OUTSIDE MODEL	opt-in comp$\to$R128$\to$limiter; pure render byte-identical by default
u=0 dynamic range / near-silence	OPEN FINDING	wide per-bar mass + global peak-normalize; no extrinsic AGC (structure-deleting class)

8 Conclusion

ETS synthesizes music by minimizing a single fixed free-energy functional F whose low-energy configurations are real tracks, with a causal writer that commits one bar at a time and control applied as an exponential tilt whose gains are derived from equilibrium fluctuations. Its output is a sequence of real source units with complete provenance rather than a synthesized waveform. Its second and more general contribution is a construction method: a single authority (F), a two-state invariant manifest with executable checks tested to fail on planted violations, an adversarial auditor, and an append-only decision log in which a caught fidelity breach and a failed dissociation gate are recorded as results. The limitations listed here — 14 of 15 invariants enforced, two control lanes disarmed, one uncertified realization step, a failed G2, and an unbuilt planner — are reported as first-class results rather than omitted.

All figures are from the committed sources named inline (`ets-spec-v0.md`, `ets-connector-v0.md`, `REGISTRY.jsonl`) and the results artifacts `g0_results.json`, `g1_results.json`, `g2_results.json`, `training_results.json`, `latency_desktop.json`, and `ets/calibration/sigma_phi.json`. This document is post-v1 documentation; it modifies no source and affects no workflow.

References

[1] Y. LeCun, S. Chopra, R. Hadsell, M. Ranzato, and F.-J. Huang. A tutorial on energy-based learning. In *Predicting Structured Data*. MIT Press, 2006.

[2] Y. Du and I. Mordatch. Implicit generation and generalization in energy-based models. *arXiv:1903.08689*, 2019.

[3] M. Gutmann and A. Hyvärinen. Noise-contrastive estimation: a new estimation principle for unnormalized statistical models. In *Proc. AISTATS* (PMLR 9), pp. 297–304, 2010.

[4] M. Cuturi. Sinkhorn distances: lightspeed computation of optimal transportation distances. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2013. *arXiv:1306.0895*.

[5] G. Peyré and M. Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6), 2019. *arXiv:1803.00567*.

[6] F. Mémoli. Gromov–Wasserstein distances and the metric approach to object matching. *Foundations of Computational Mathematics*, 11(4):417–487, 2011.

[7] G. Peyré, M. Cuturi, and J. Solomon. Gromov–Wasserstein averaging of kernel and distance matrices. In *Proc. ICML* (PMLR 48), pp. 2664–2672, 2016.

[8] T. Henderson and J. Solomon. Audio transport: a generalized portamento via optimal transport. In *Proc. Int. Conf. on Digital Audio Effects (DAFx)*, 2019. *arXiv:1906.06763*.

[9] D. Schwarz, G. Beller, B. Verbrugghe, and S. Britton. Real-time corpus-based concatenative synthesis with CataRT. In *Proc. Int. Conf. on Digital Audio Effects (DAFx)*, 2006.

[10] D. Schwarz. Corpus-based concatenative synthesis. *IEEE Signal Processing Magazine*, 24(2):92–104, 2007.

[11] M. E. P. Davies, P. Hamel, K. Yoshii, and M. Goto. AutoMashUpper: automatic creation of multi-song music mashups. *IEEE/ACM Trans. Audio, Speech, and Language Processing*, 22(12), 2014.

[12] J. Engel, L. Hantrakul, C. Gu, and A. Roberts. DDSP: differentiable digital signal processing. In *Int. Conf. on Learning Representations (ICLR)*, 2020. *arXiv:2001.04643*.

[13] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi. SoundStream: an end-to-end neural audio codec. *arXiv:2107.03312*, 2021.

[14] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi. High fidelity neural audio compression. *arXiv:2210.13438*, 2022.

[15] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, and A. Défossez. Simple and controllable music generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. *arXiv:2306.05284*.